Causal Inference Through the Lens of Probabilistic Programming

PyCon DE & PyData 2026

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Outline

What is Causal Inference?

- **Goal:** Estimate the effect of a treatment T on an outcome Y.
- **Potential outcomes** (Rubin, 1974): For each unit i, there exist two potential outcomes:
  - $Y_i(1)$: outcome if treated
  - $Y_i(0)$: outcome if not treated
- **Individual Treatment Effect:** $\tau_i = Y_i(1) - Y_i(0)$
- **Average Treatment Effect (ATE):**

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)]$$

⚠ **The Fundamental Problem of Causal Inference**

We can **never observe both** $Y_i(1)$ and $Y_i(0)$ for the same unit. One potential outcome is always **missing** (counterfactual).

$$\text{Observed: } Y_i = T_i \cdot Y_i(1) + (1 - T_i) \cdot Y_i(0)$$

This is why naive comparisons (treated vs. untreated) can be **biased**: the groups may differ systematically due to **confounders**.

What is Bayesian Statistics?

Bayes' theorem: Update beliefs about parameters θ after observing data y:

$$\underbrace{p(\theta \mid y)}_{\text{posterior}} \propto \underbrace{p(y \mid \theta)}_{\text{likelihood}} \cdot \underbrace{p(\theta)}_{\text{prior}}$$

- **Prior** $p(\theta)$: encodes domain knowledge before seeing data.
- **Likelihood** $p(y \mid \theta)$: how probable the data is given the parameters.
- **Posterior** $p(\theta \mid y)$: updated beliefs after observing data.

The PPL bridge

In a generative graphical model, every distribution **starts as a prior** over the data-generating process. Once we **condition on observed data**, those priors become **likelihoods**. A causal model and a Bayesian model are the **same object**, viewed before and after observing data.

Probabilistic Programming Languages (PPLs)

PPLs let you write generative models as code and automate posterior inference. Examples: PyMC, NumPyro, Stan.

Why PPLs for Causal Inference?

- **Model = DAG:** Writing the generative process forces you to explicitly state your causal assumptions: this alone is valuable!
- **Uncertainty for free:** Posterior distributions over causal effects.
- **Prior Modeling:** Setting priors to reflect our domain knowledge.
- **Flexible modeling:** Swap likelihoods (Normal $\rightarrow$ Gamma), add hierarchical structure (Mundlak), or latent variables for unobserved confounders (CEVAE).
- **do-operator:** Intervene on the generative model for counterfactual reasoning.
- **Fast inference:** Modern MCMC (NUTS) and SVI make posterior computation scalable and reliable.

This **does not** imply that Bayesian methods are better than frequentist methods! For every framework, you need to know the assumptions and the limitations.

PPLs do not solve **identification**! They make **estimation** under causal assumptions.

Bayesian A/B Testing

A/B testing is the **gold standard of causal inference**: random assignment makes treatment independent of all confounders, observed and unobserved. The hard part shifts from **identification** to **estimation under uncertainty**.

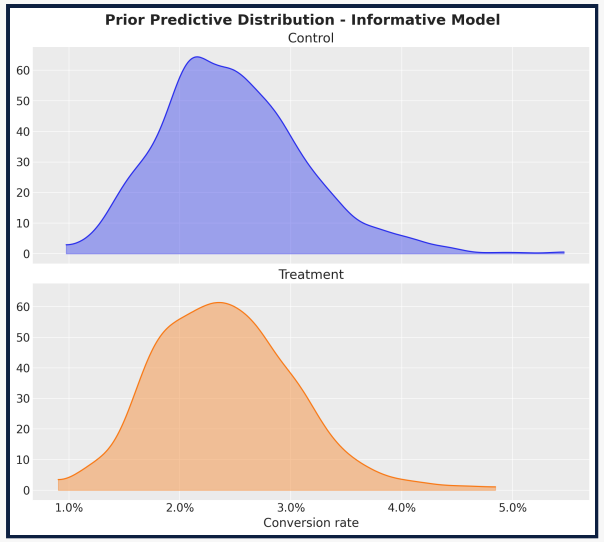
```
with pm.Model() as non_informative_model:
    conversion_rate_control = pm.Uniform(
      "conversion_rate_control", lower=0, upper=1
    )
    conversion_rate_treatment = pm.Uniform(
        "conversion_rate_treatment", lower=0, upper=1
    )
    relative_lift = pm.Deterministic(
        "relative_lift",
        conversion_rate_treatment / conversion_rate_control - 1,
    )
```

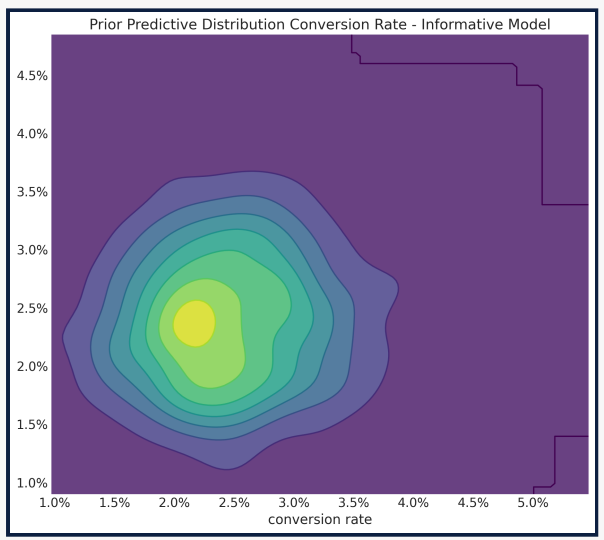
Uninformative Priors: Implications

Uninformative Priors yield implausible values for the relative lift.

Setting uninformative priors for the conversion rates yields implausible values for the relative lifts as this is a ratio of two random variables, so it can blow up to infinity or zero.

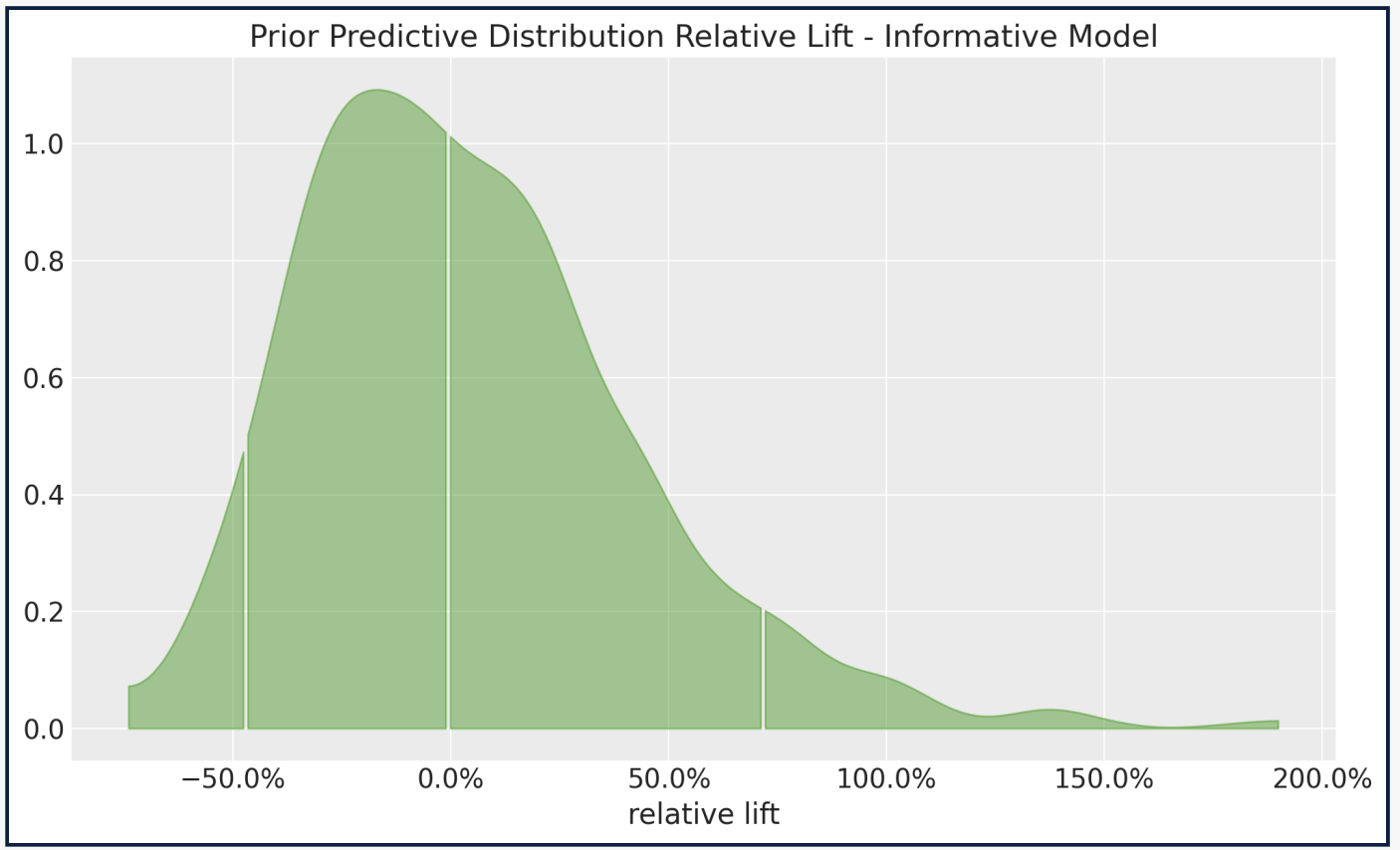
Informative Priors: Implications





⚠ Informative Priors

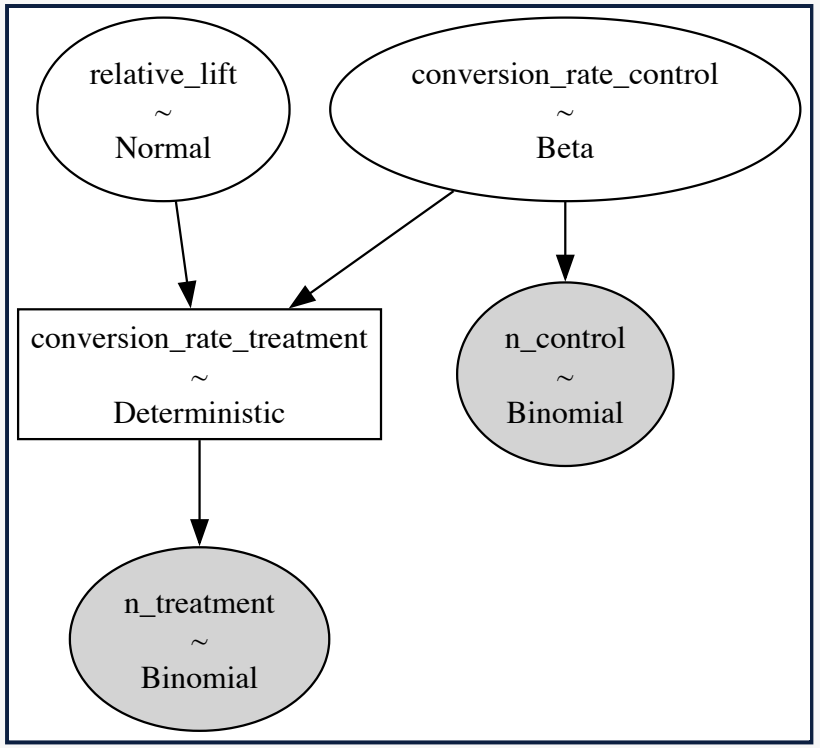
Even if we set informative priors for the conversion rates, the relative lift range is still unreasonably wide.

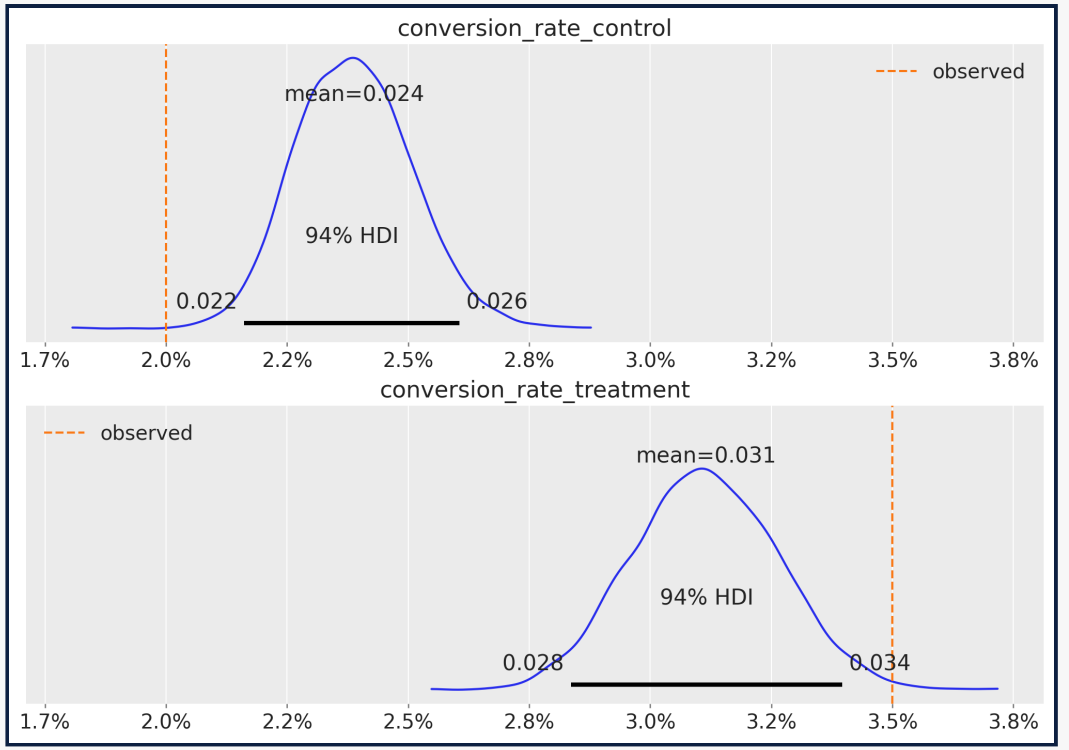


Correlated Priors: Implications

Inference with Correlated Priors

```
with correlated_model:
    pm.Binomial(
      "n_control", n=n, p=conversion_rate_control, observed=n_c
    )
    pm.Binomial(
      "n_treatment", n=n, p=conversion_rate_treatment, observed=n_t
    )
```


relative_lift
~
Normal
conversion_rate_control
~
Beta
conversion_rate_treatment
~
Deterministic
n_control
~
Binomial
n_treatment
~
Binomial


conversion_rate_control
observed
mean=0.024
94% HDI
0.022
0.026
1.7%
2.0%
2.2%
2.5%
2.8%
3.0%
3.2%
3.5%
3.8%
conversion_rate_treatment
observed
mean=0.031
94% HDI
0.028
0.034
1.7%
2.0%
2.2%
2.5%
2.8%
3.0%
3.2%
3.5%
3.8%

Bayesian Power Analysis

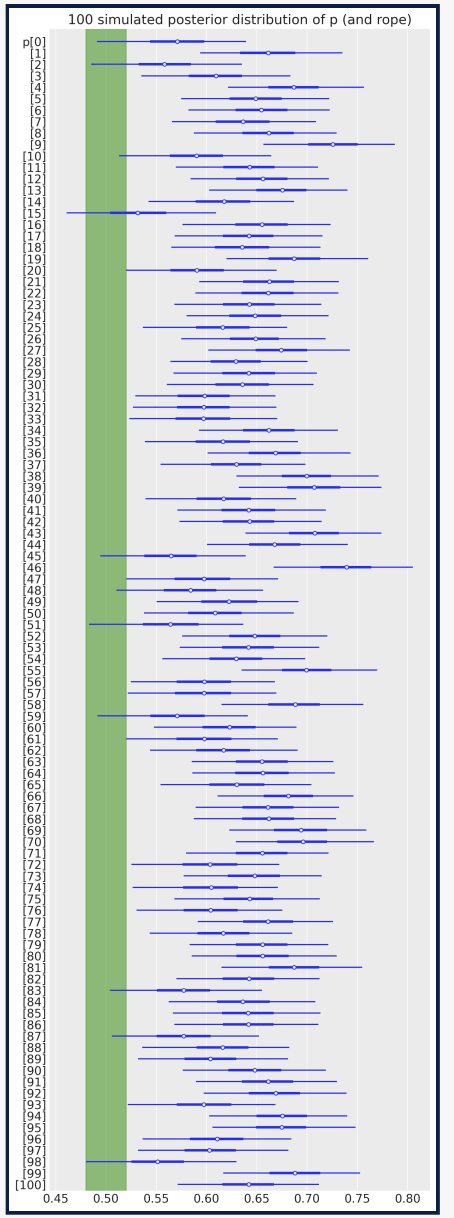
HDI + ROPE Framework

The Bayesian New Statistics: Hypothesis testing, estimation, meta-analysis, and power analysis from a Bayesian perspective

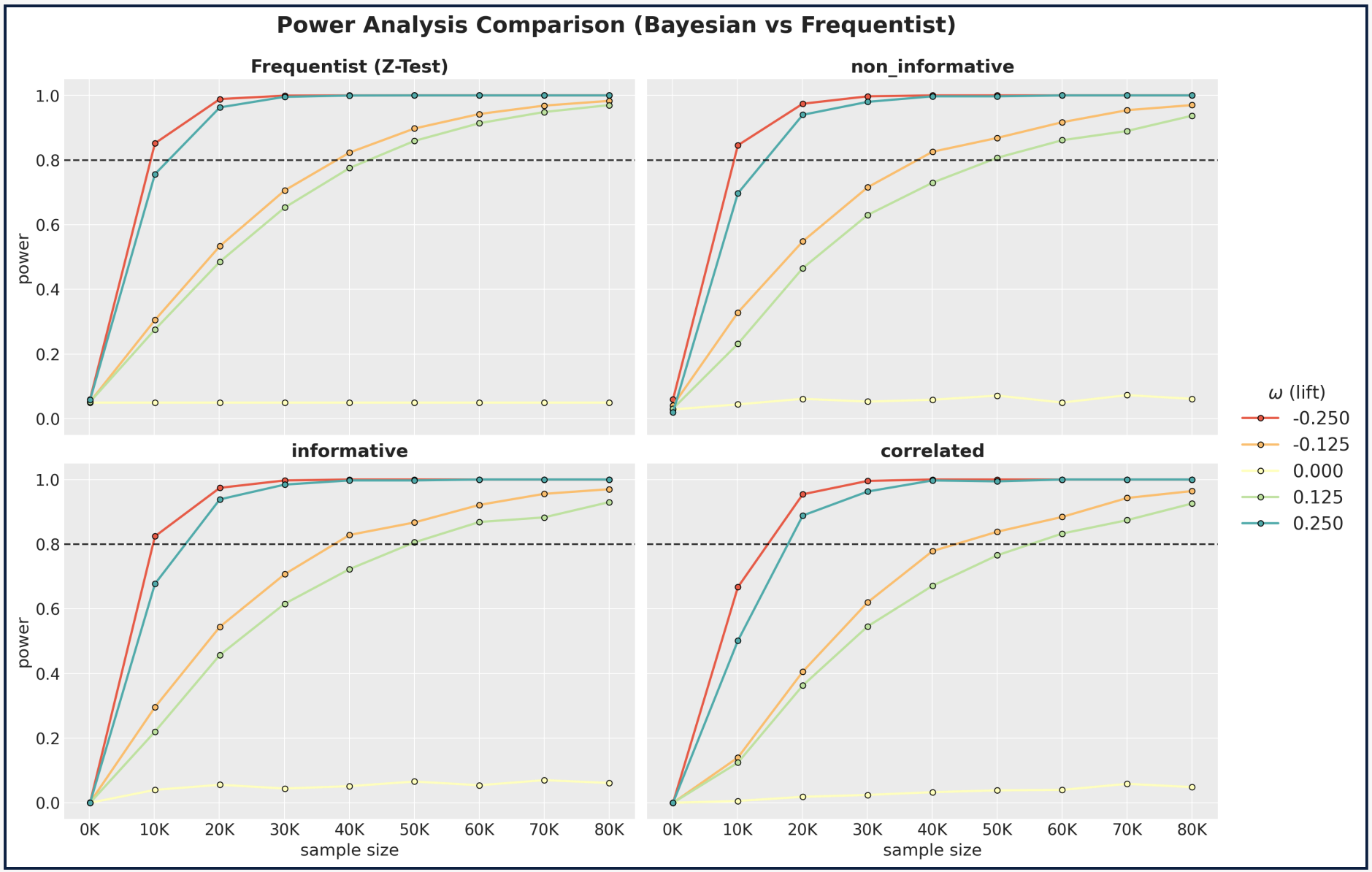
Declare significance when the 94% HDI of the posterior lift **excludes the ROPE** (Region of Practical Equivalence) around zero.

Five-step process:

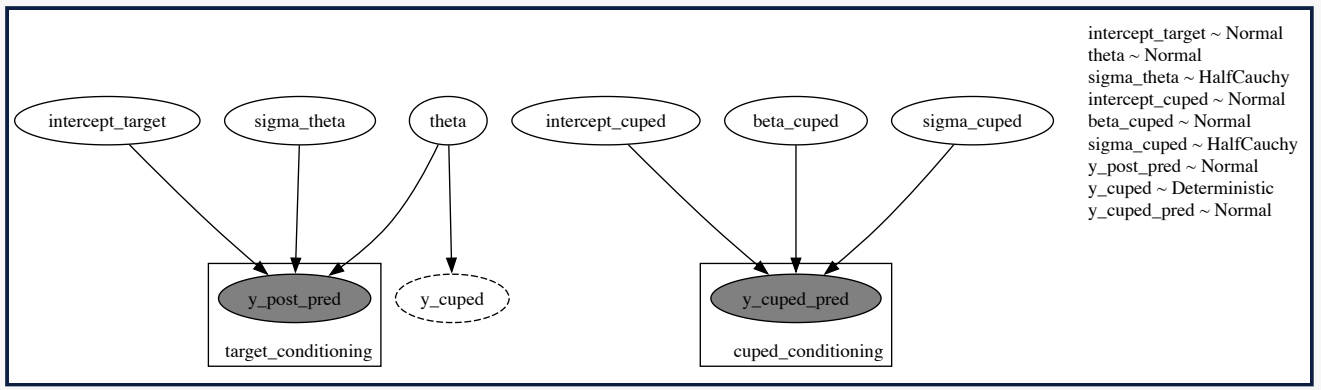
1. Generate parameter values from a hypothetical distribution
2. Simulate data samples using the generative model
3. Compute posterior estimates with Bayesian analysis for various sample sizes
4. Assess whether the 94% HDI excludes the ROPE
5. Repeat to approximate statistical power



Power Curves



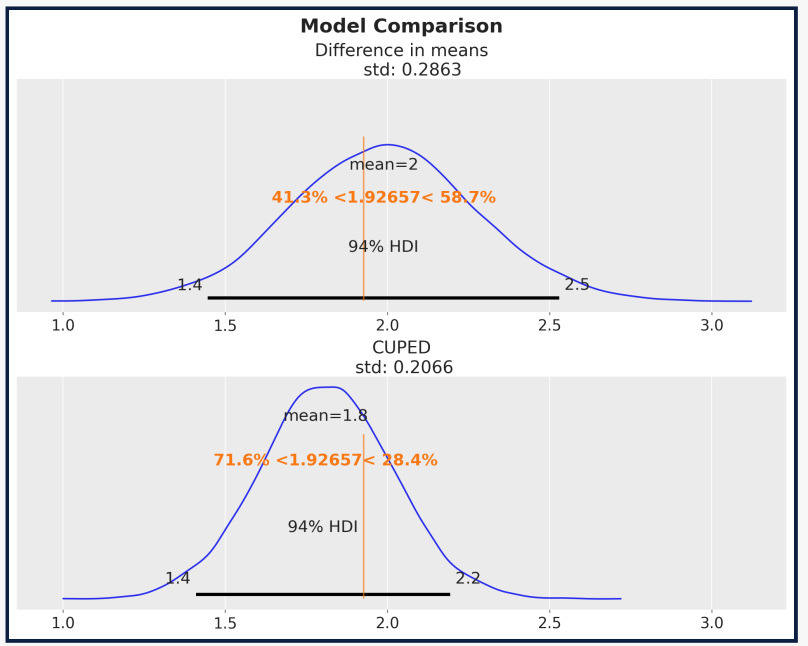
Bayesian CUPED



CUPED: Use pre-treatment outcome as a covariate to **reduce posterior variance of the causal effect estimate**.

1. Regress y_{post} on y_{pre} and estimate the θ coefficient.
2. Compute $y_{\text{cuped},i} = y_{\text{post},i} - \theta\left(y_{\text{pre},i} - \bar{y}_{\text{pre}}\right)$ for each unit i.
3. Compute the difference in means of y_{cuped} between treatment and control group.

$$y_{\text{cuped}} = \alpha + \beta_{\text{cuped}} \times \text{treatment} + \varepsilon$$

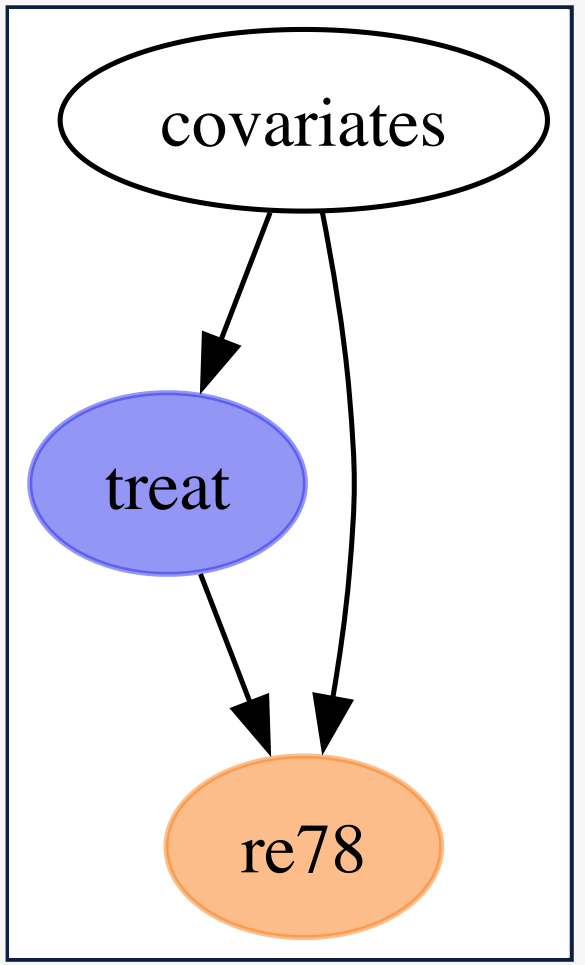


Confounders: An Example

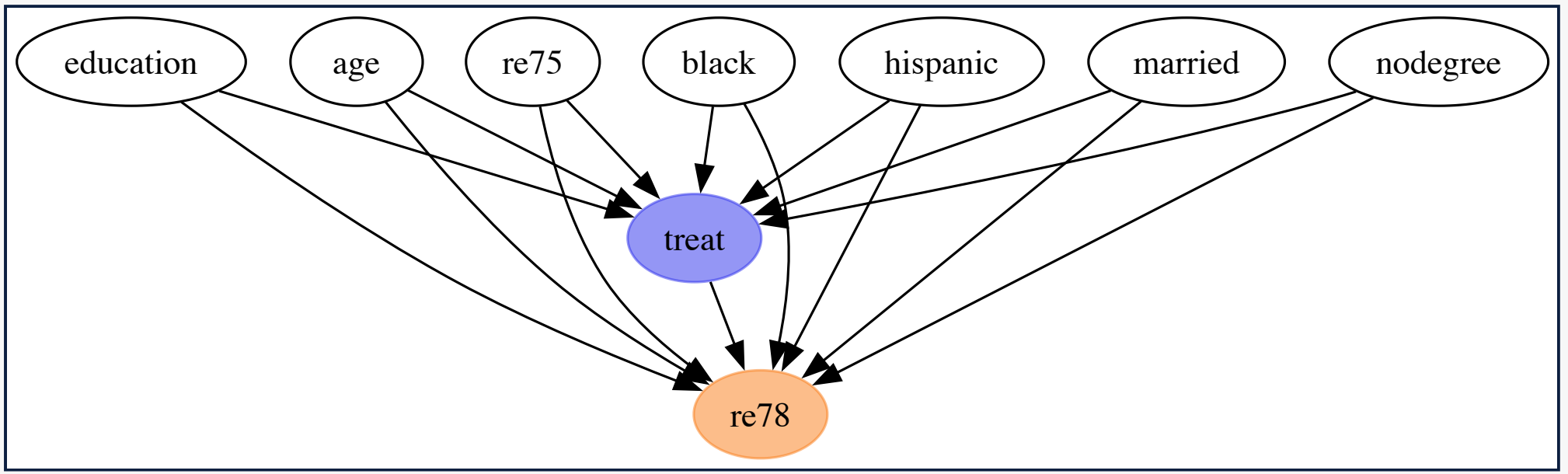
The Lalonde Dataset: Job training program (National Supported Work, 1970s)

Question: Does job training increase earnings?

- **Treatment:** Participation in a job training program
- **Outcome:** Earnings in 1978 (`re78`)
- **Covariates:** Education, age, prior earnings (`re75`), race/ethnicity, marital status, degree
- **Naive comparison** (difference in means): treated earned **$635 less** than untreated
- But treatment was **not** randomly assigned…
- **Confounders** create a spurious association!



The Causal DAG



Backdoor criterion: block all backdoor paths from treatment to outcome to identify the causal effect.

ATE Estimation Strategies

Two ways to estimate the ATE:

(1) Read the estimate from a regression coefficient:

Fit a linear model of the form

$$\text{earnings} = \alpha + \beta_{\text{treat}}\text{treat} + \beta_{\text{covariates}}\text{covariates} + \varepsilon$$

and identify the estimate of β_{treat} as the ATE.

(2) Use the do operator for counterfactual computation:

$$\text{ATE} = \mathbb{E}[\text{earnings} \mid \text{do}(\text{treat} = 1)] - \mathbb{E}[\text{earnings} \mid \text{do}(\text{treat} = 0)]$$

Pearl's do operator and counterfactuals

The do operator implements Pearl's intervention:

$$P(Y \mid \text{do}(T = t))$$

It **cuts** incoming edges to the treatment node, simulating a randomized experiment.

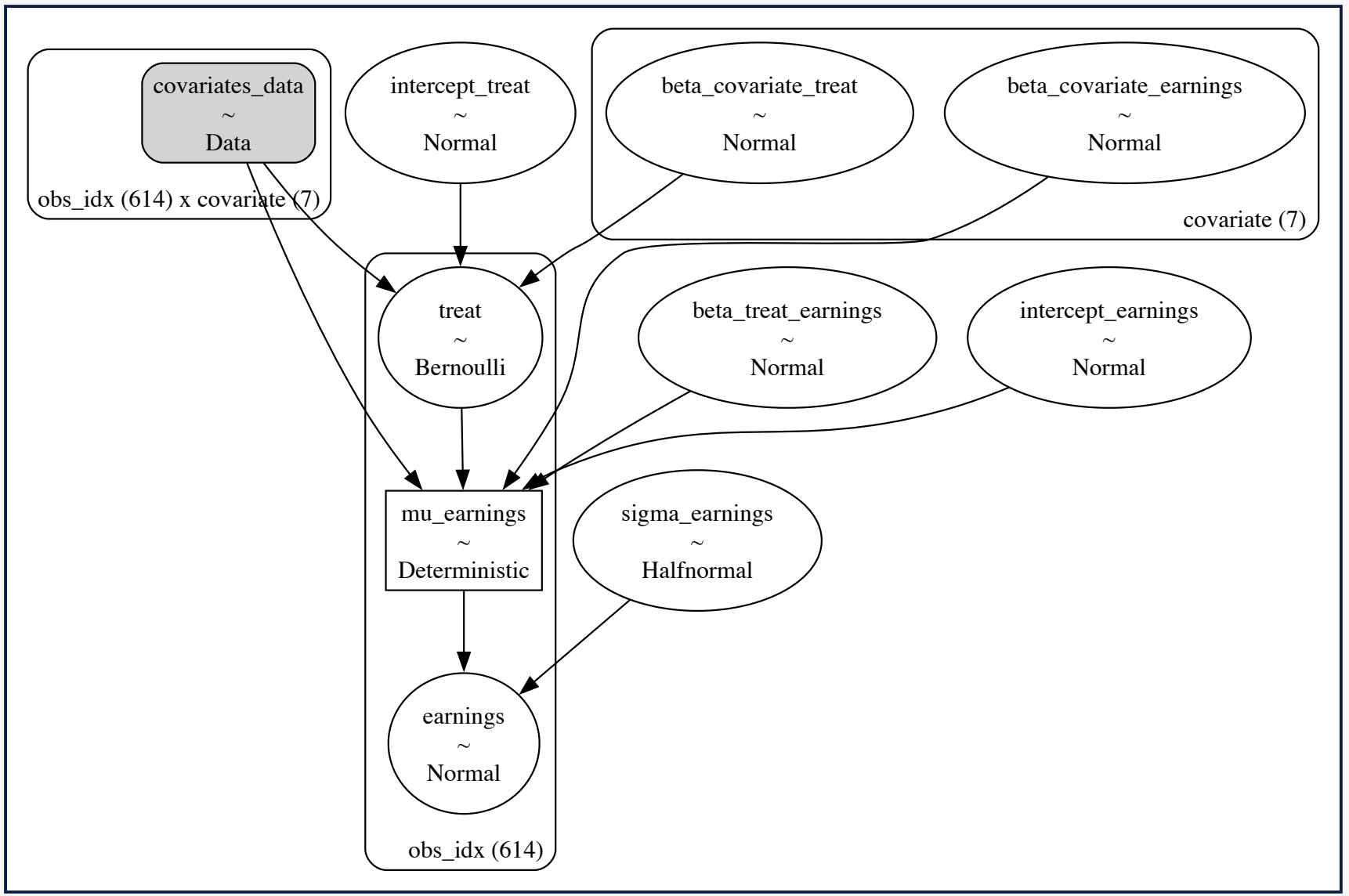
PyMC Model: Treatment Sub-model

```python
with pm.Model(coords=coords) as earnings_model:
    # --- Data ---
    covariates_data = pm.Data(
        "covariates_data", covariates_obs, dims=("obs_idx", "covariate")
    )

    # --- Priors ---
    intercept_treat = pm.Normal("intercept_treat", mu=0, sigma=10)
    beta_covariate_treat = pm.Normal(
        "beta_covariate_treat", mu=0, sigma=1, dims=("covariate",)
    )

    # --- Parametrization ---
    logit_p_treat = intercept_treat + pm.math.dot(
        covariates_data, beta_covariate_treat
    )
    p_treat = pm.math.sigmoid(logit_p_treat)

    # --- Likelihood ---
    treat = pm.Bernoulli("treat", p=p_treat, dims=("obs_idx",))
```

PyMC Model: Earnings Sub-model

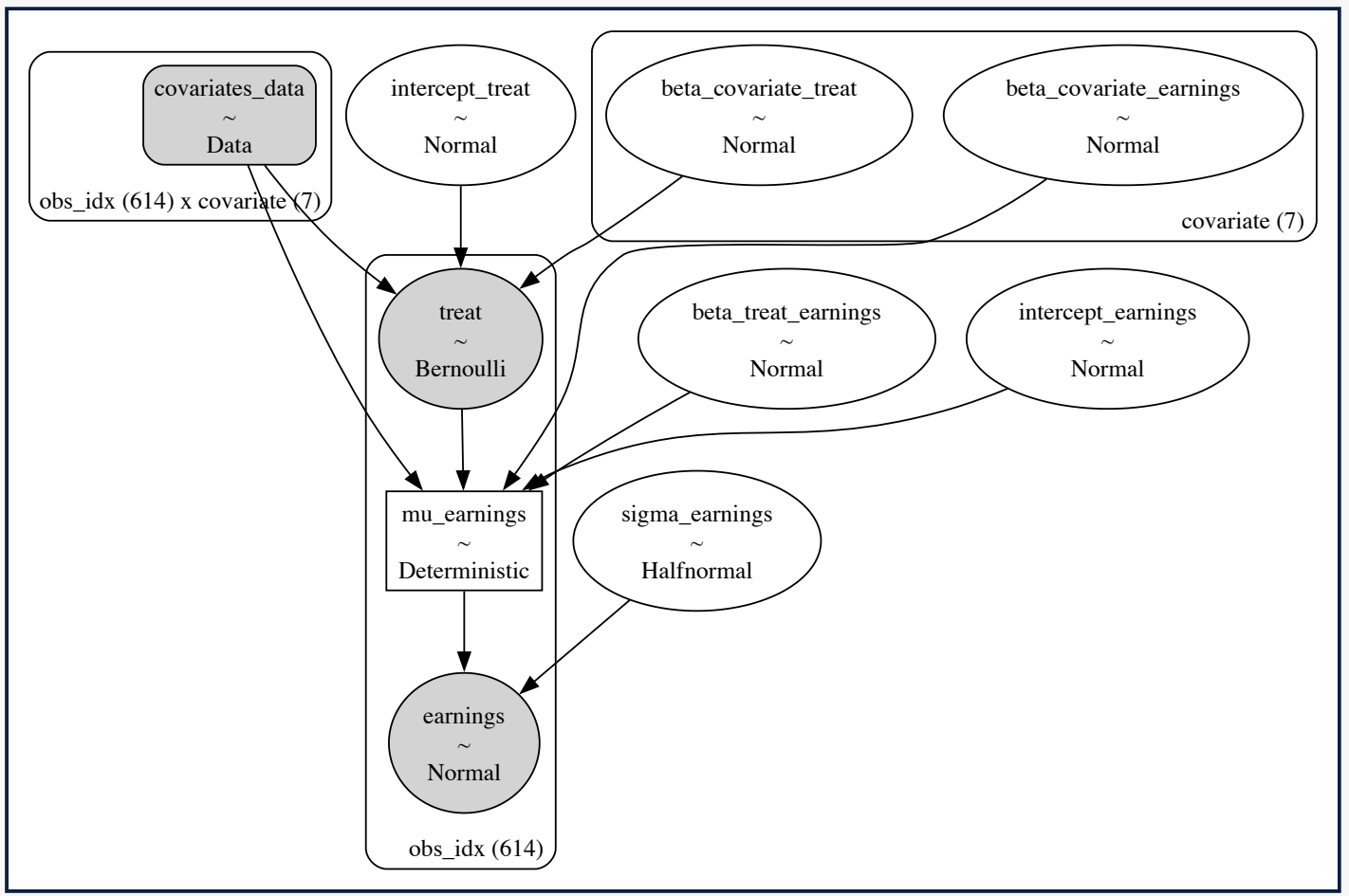
```
    # --- Priors ---
    intercept_earnings = pm.Normal("intercept_earnings", mu=0, sigma=10)
    beta_treat_earnings = pm.Normal("beta_treat_earnings", mu=0, sigma=1)
    beta_covariate_earnings = pm.Normal(
        "beta_covariate_earnings", mu=0, sigma=1, dims=("covariate",)
    )
    sigma_earnings = pm.HalfNormal("sigma_earnings", sigma=10.0)

    # --- Parametrization ---
    mu_earnings = pm.Deterministic(
        "mu_earnings",
        intercept_earnings + beta_treat_earnings * treat
            + pm.math.dot(covariates_data, beta_covariate_earnings),
        dims=("obs_idx",),
    )

    # --- Likelihood ---
    pm.Normal(
      "earnings", mu=mu_earnings, sigma=sigma_earnings, dims=("obs_idx",)
    )
```

PyMC Model: Generative Process

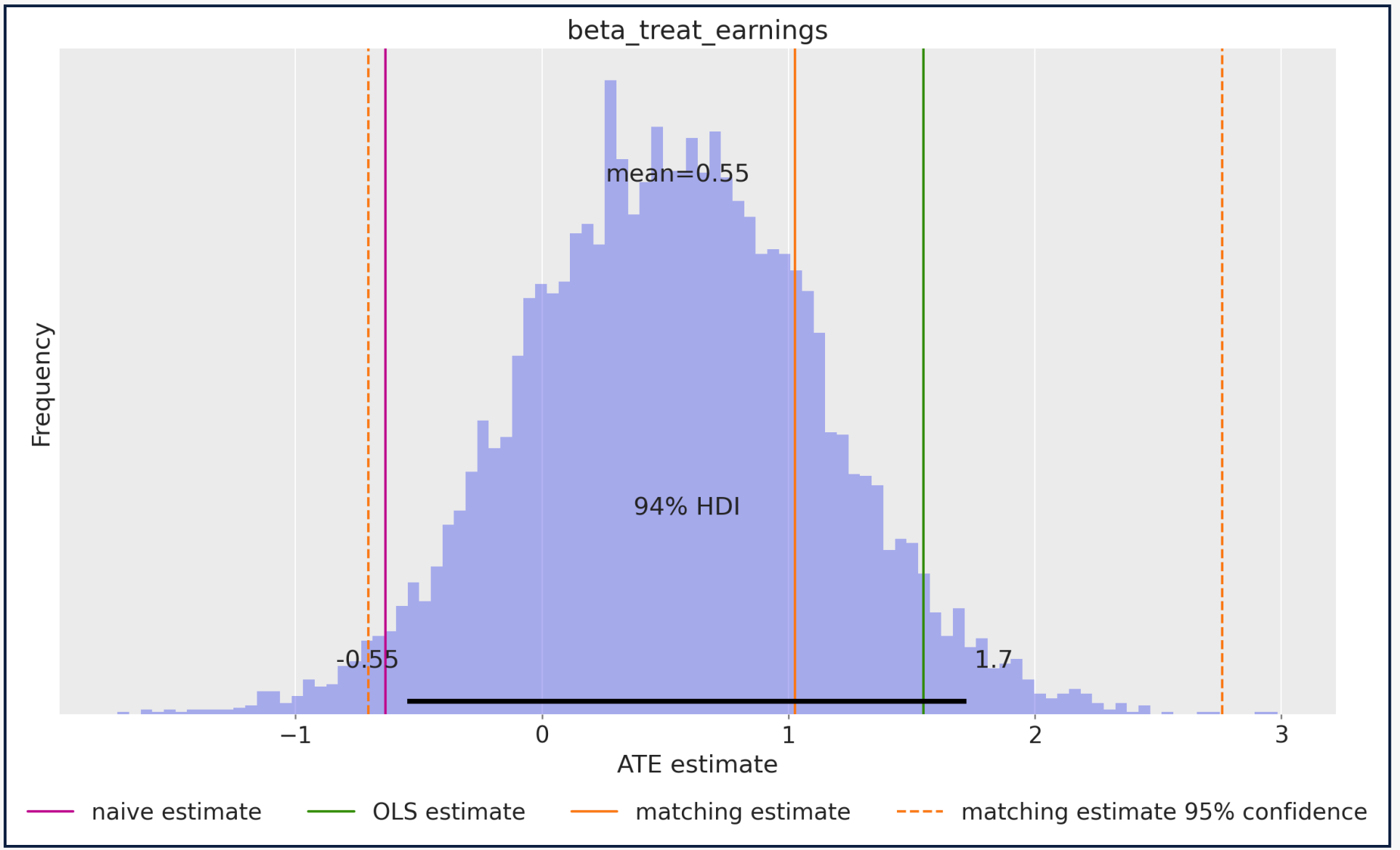


Conditioning on Observed Data

```
1 conditioned_earnings_model = observe(
2     earnings_model, {"treat": training_obs, "earnings": earnings_obs}
3 )
```



ATE from the Regression Coefficient



The do Operator

The do operator implements Pearl's intervention:

$$P(Y \mid \text{do}(T = t))$$

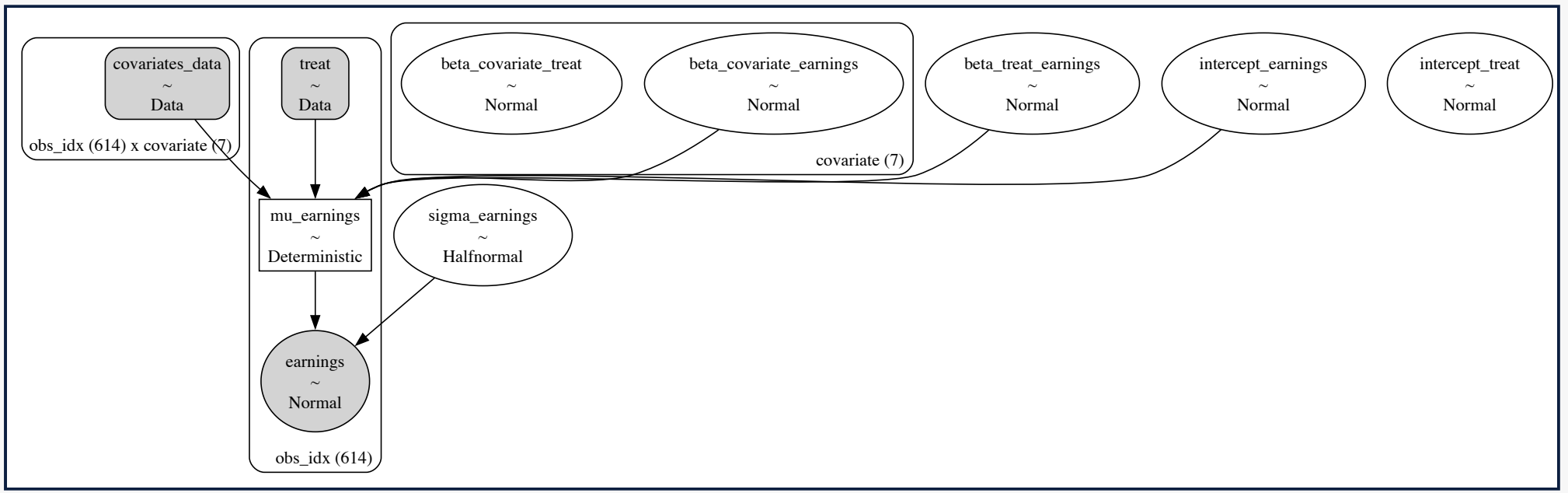
It **cuts** incoming edges to the treatment node, simulating a randomized experiment.

We can use the do operator to compute the ATE:

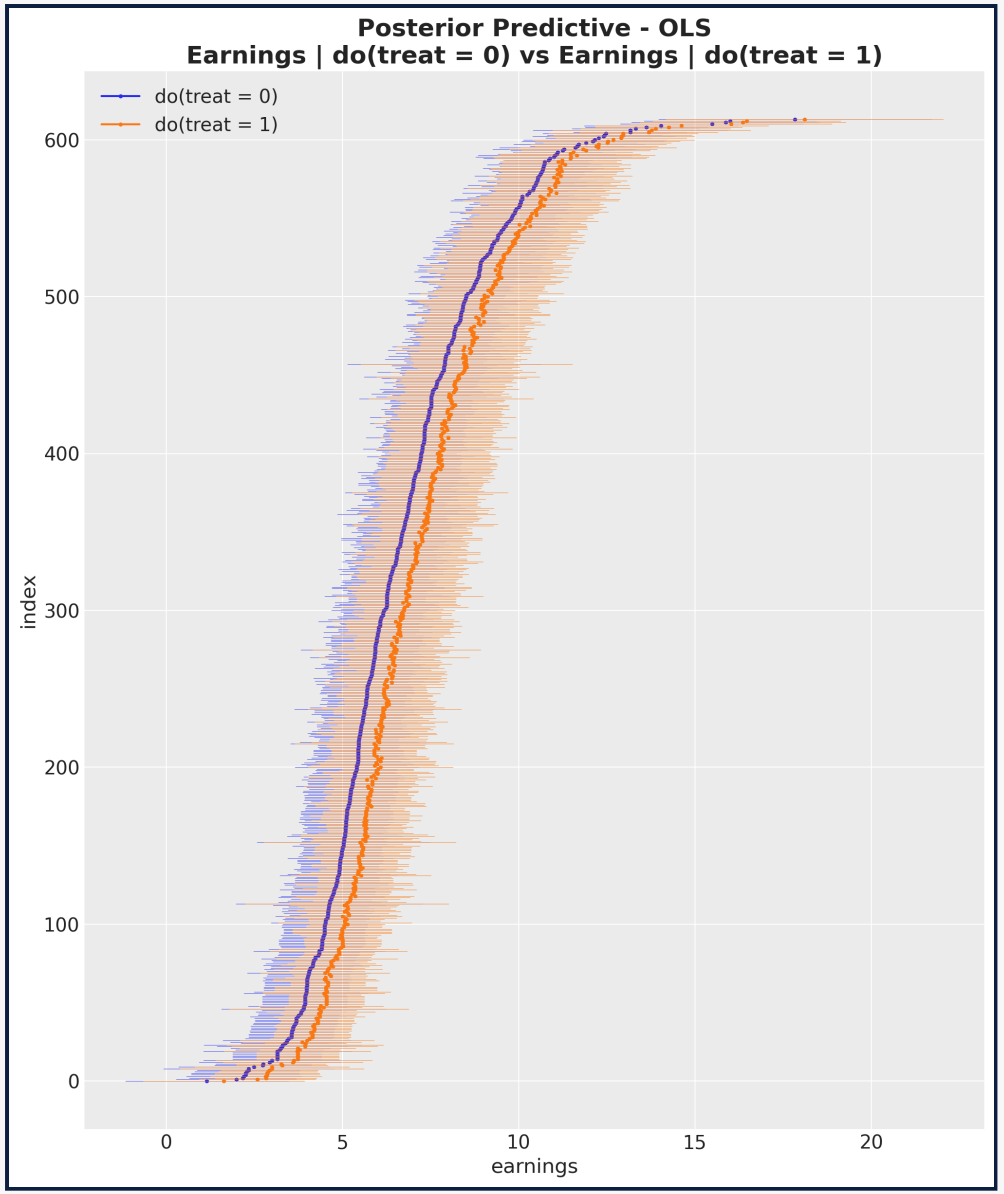
$$ATE = \mathbb{E}[Y \mid \text{do}(T = 1)] - \mathbb{E}[Y \mid \text{do}(T = 0)]$$

Do-Operator Resulting Model Graph

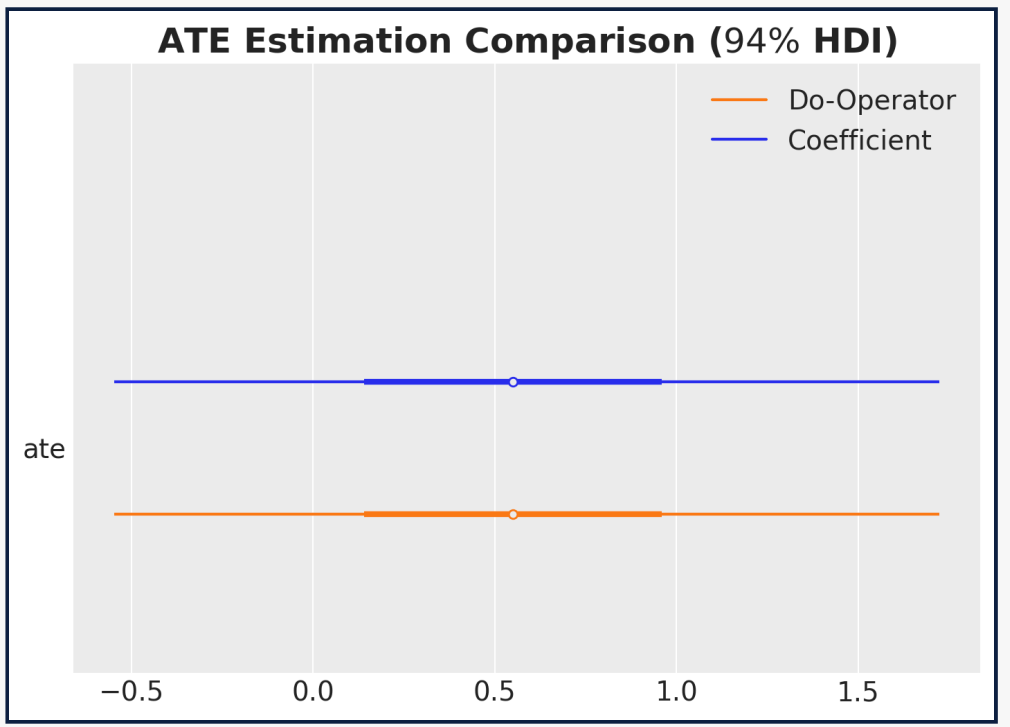
```python
from pymc.model.transform.conditioning import do
# Counterfactual interventions
do_0_model = do(conditioned_earnings_model, {"treat": np.zeros(n_obs)})
do_1_model = do(conditioned_earnings_model, {"treat": np.ones(n_obs)})
```



Individual Counterfactual Predictions

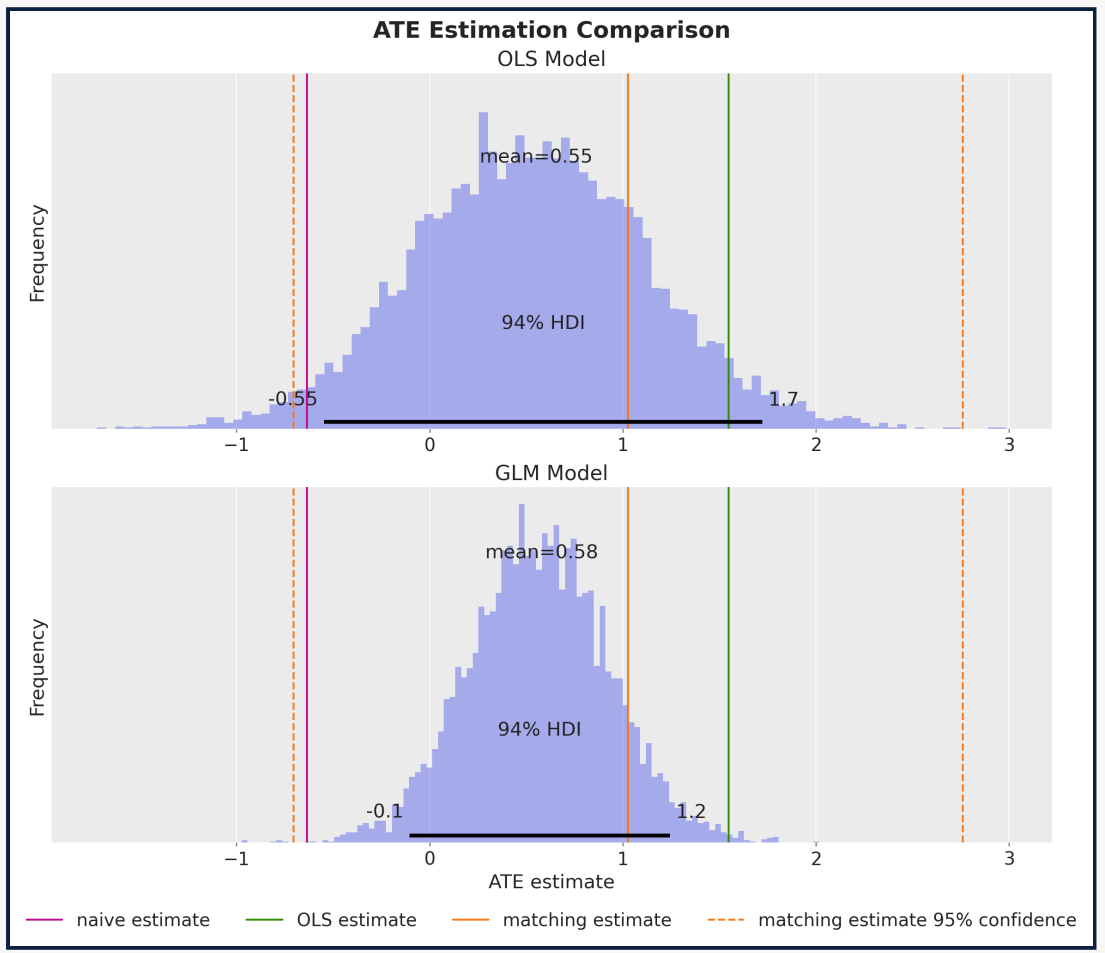


Computing the ATE using the do operator yields the same result as the regression coefficient approach.



A more meaningful likelihood

Gamma Generalized Linear Model

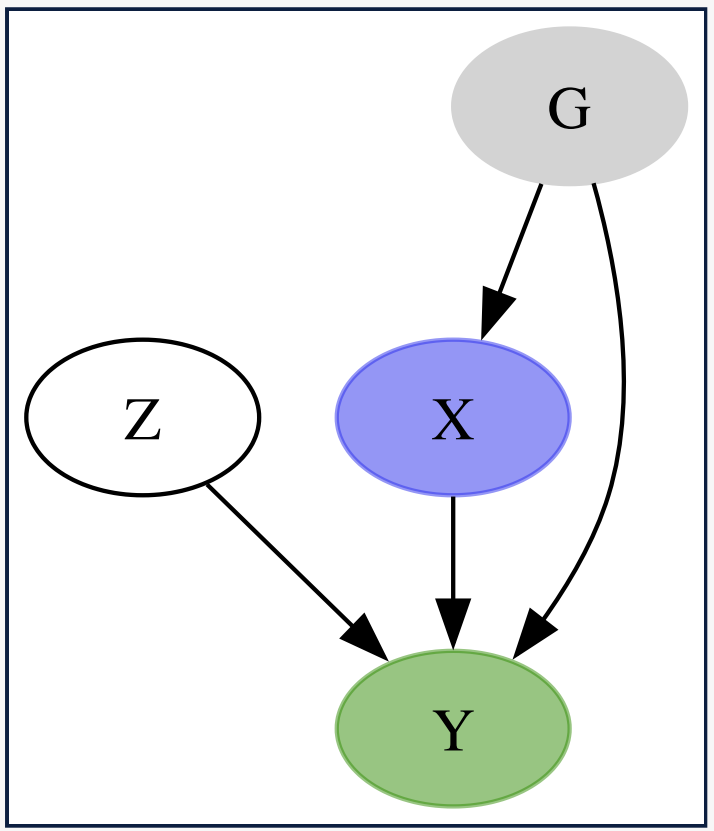


- Earnings are **non-negative and right-skewed** — Normal puts mass on impossible values.
- **One-line model change:** `pm.Normal(...)` → `pm.Gamma(...)` with a log link.
- ATE on the original scale comes for free via the do-operator.

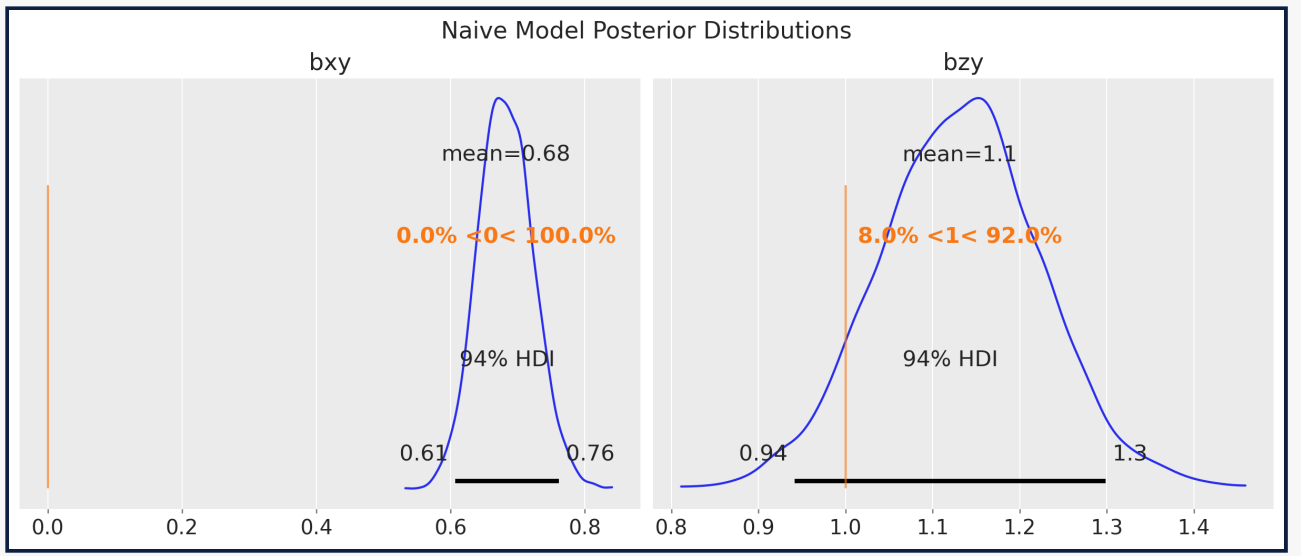
Why the do-operator now matters: with a non-linear link, the regression coefficient is **not** the ATE. The do-operator handles non-linearity, individual counterfactuals, and joint interventions on multiple variables.

Fixed & Random Effects

Group-Level Confounding



Problem: Synthetic dataset where the true causal effect of X on Y is **zero**. We have **group-level confounding** through group-level effect U_G.



Taken from Richard McElreath's Statistical Rethinking (2026 Lectures).

Five Estimation Strategies

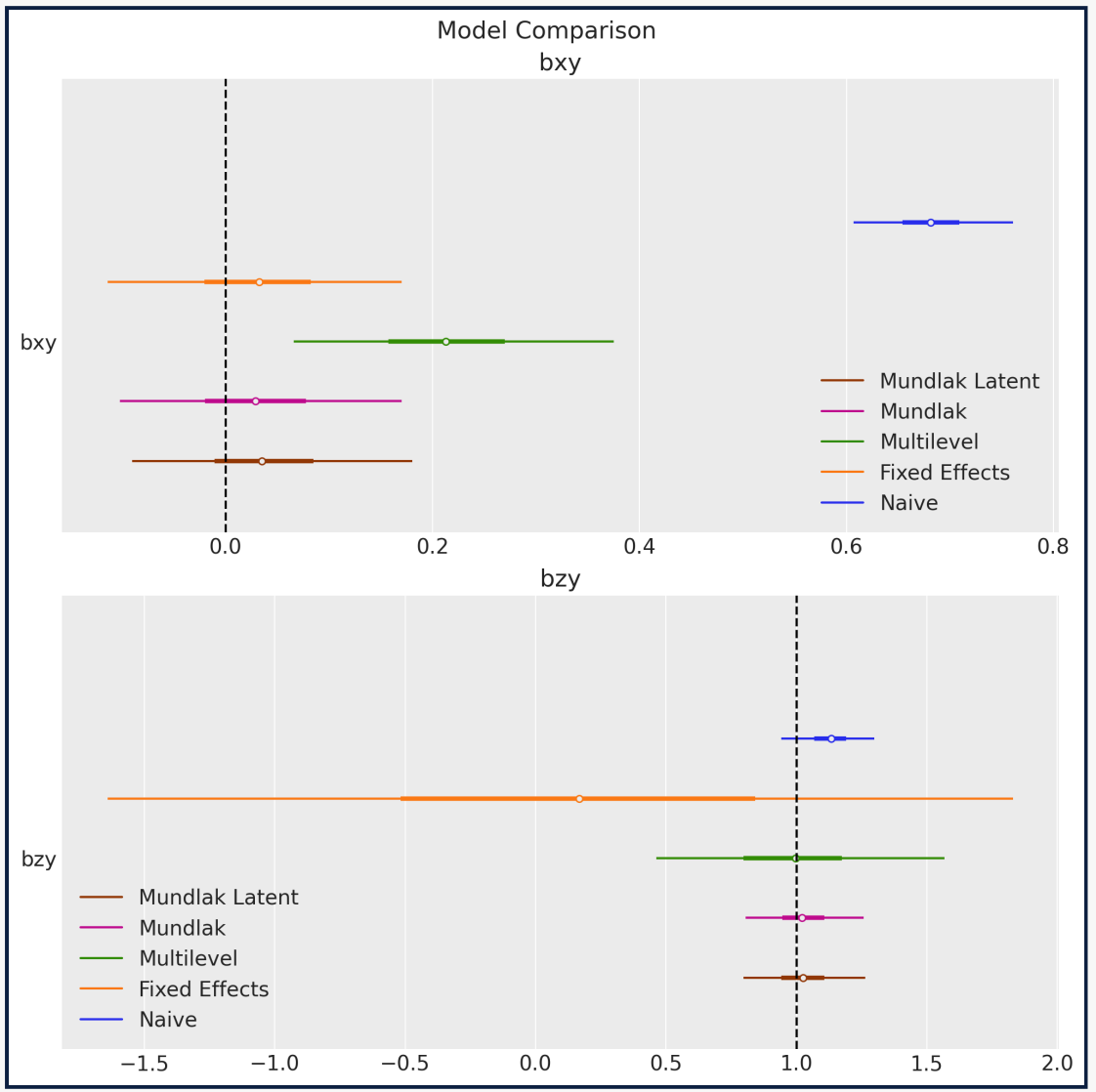
Model Structure

Model	Approach	Recovers true effect?
Naive	Ignores groups entirely	No (biased)
Fixed Effects	Group-specific intercepts	Yes (high variance)
Multilevel	Hierarchical priors on groups	No (confounded)
Mundlak	Adds group means of X	Yes (efficient)
Mundlak Latent	Latent group variable	Yes (best by LOO)

$$y = \beta_{xy}x + \beta_{zy}z + ? + \varepsilon$$

Key insight (Mundlak trick): Including group-level means as predictors absorbs the between-group confounding while preserving partial pooling benefits.

Results: The Mundlak Latent Model



- Fixed effects

```
y ~ 0 + x + z + C(g)
```

- Multilevel

```
y ~ x + z + (1 | g)
```

- Mundlak

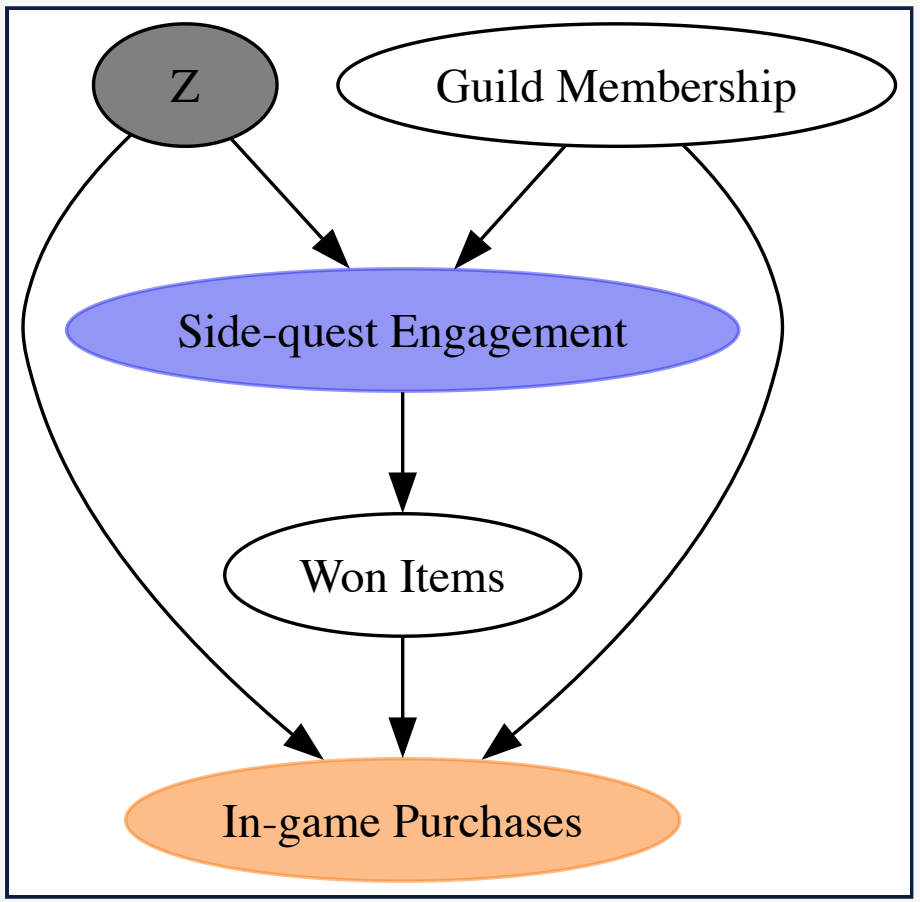
```
y ~ x + z + (1 | g) + x_bar
```

- Mundlak Latent

$$u_{[g]} \sim \text{Normal}(\mu_u, \sigma_u)$$
$$x = \alpha_x + \beta_{ux} u_{[g]} + \varepsilon_x$$
$$y = \alpha_{y,[g]} + \beta_{xy} x + \beta_{zy} z + \beta_{yg} u_{[g]} +$$

Latent Variable Modeling

Unobserved Confounders

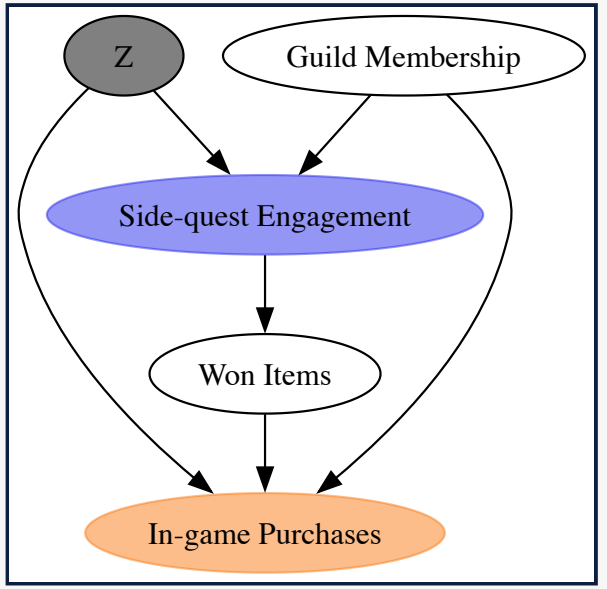


- Online game dataset: `Does side-quest engagement increase in-game purchases?`
- **Unobserved confounder** Z (player motivation) affects both treatment and outcome.
- Standard backdoor adjustment is **not possible**.
- A solution: **CEVAE** framework (Louizos et al., NeurIPS 2017): learn a latent representation of Z.

Taken from Robert O. Ness's book: Causal AI.

CEVAE Architecture

- **Model = Decoder (generative process):** Follows the causal ordering:
  1. Z (player motivation) $\sim \text{Normal}(0, 1)$
  2. Engagement (treatment) $\sim \text{Bernoulli}(f_\theta(\text{guild membership}, Z))$
  3. Purchases (outcome) $\sim \text{Normal}(g_\theta(\text{won items}, \text{guild membership}, Z))$
- **Guide = Encoder (recognition network):** Flax (NNX) neural network that maps observed data $(\text{guild membership}, \text{engagement}, \text{purchases})$ to the approximate posterior $q_\phi(Z \mid \cdot) = \text{Normal}(\mu_\phi, \sigma_\phi)$
- **Inference:** SVI maximizes the ELBO, jointly training the decoder parameters θ and encoder parameters ϕ (NumPyro)

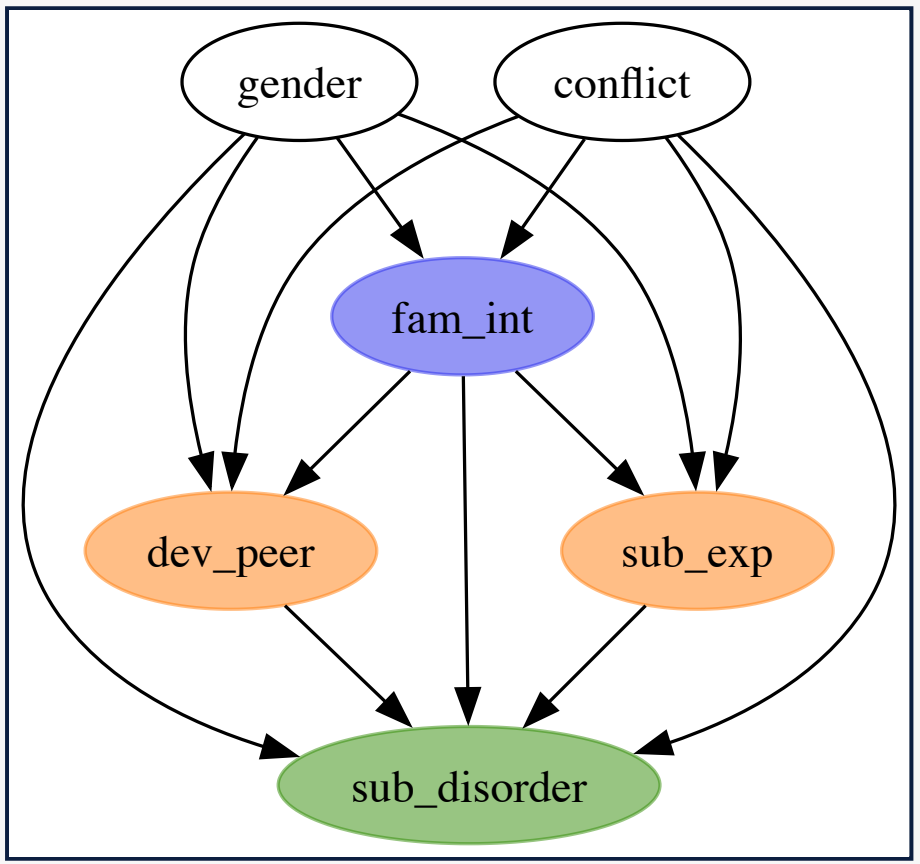


Simulation Study

Here is a simpler parameter recovery study for the CEVAE: CATE Estimation with Causal Effect Variational Autoencoders

Mediation Analysis: (In)Direct Effects

Question: How does family intervention reduce substance use disorder? Decompose total effect into direct and indirect pathways.



- Synthetic dataset ($N = 410$) studying family intervention effects during adolescence.
- **Covariates:** `gender` (binary), `conflict` (ordinal family conflict level).
- **Treatment:** `fam_int` (family intervention program, binary).
- **Two mediators:** `dev_peer` (deviant peer engagement), `sub_exp` (drug experimentation).
- **Outcome:** `sub_disorder` (substance use disorder diagnosis, binary).

Mediation Analysis: (In)Direct Effects

Effect decomposition

Define $E_{t,t',t''}$ as the expected outcome when `fam_int` $= t$ in the outcome equation, `dev_peer` follows its $\text{do}(T = t')$ distribution, and `sub_exp` follows its $\text{do}(T = t'')$ distribution:

$$E_{t,t',t''} = \sum_{m_1,m_2} P(M_1 = m_1 \mid \text{do}(T = t')) \cdot P(M_2 = m_2 \mid \text{do}(T = t'')) \cdot P(Y = 1 \mid \text{do}(T = t, M_1 = m_1, M_2 = m_2))$$

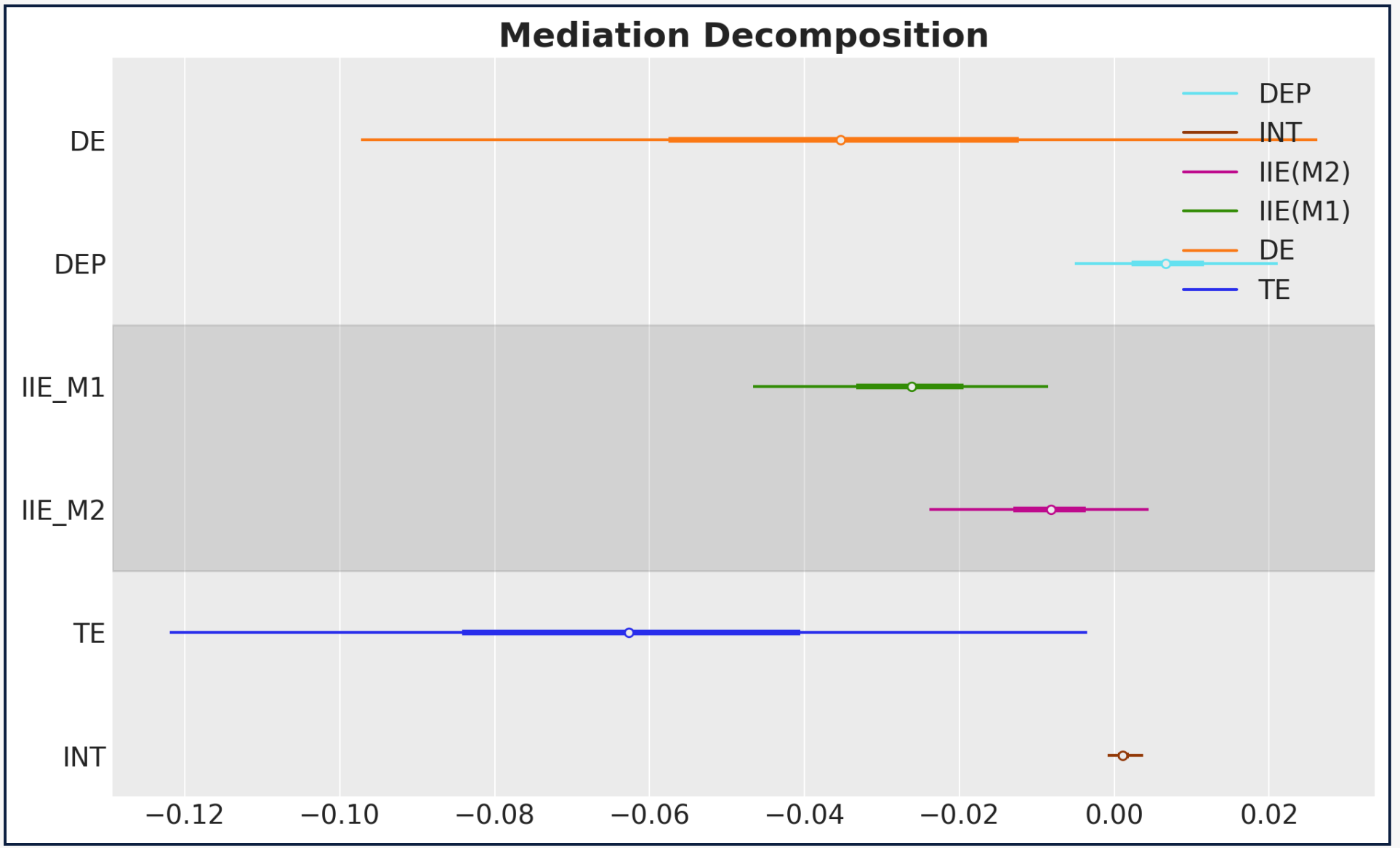
The mediators factor because there is **no edge** between `dev_peer` and `sub_exp` in the DAG.

We can compute the effects using the `do`-operator:

Effect	Formula	Interpretation
TE (Total)	$E_{1,1,1} - E_{0,0,0}$	Full treatment effect
DE (Direct)	$E_{1,0,0} - E_{0,0,0}$	`fam_int` $\rightarrow$ `sub_disorder`, mediators at control
IIE$_1$ (via `dev_peer`)	$E_{0,1,0} - E_{0,0,0}$	Shift `dev_peer` to treated, hold rest at control
IIE$_2$ (via `sub_exp`)	$E_{0,0,1} - E_{0,0,0}$	Shift `sub_exp` to treated, hold rest at control
INT (Interaction)	$E_{0,1,1} - E_{0,1,0} - E_{0,0,1} + E_{0,0,0}$	Joint mediator contribution beyond individual effects

Takeaway: With PPLs, you decompose a single treatment effect into mechanism-specific pathways — each one a posterior distribution, each one a counterfactual query against the **same model**.

Mediation Decomposition



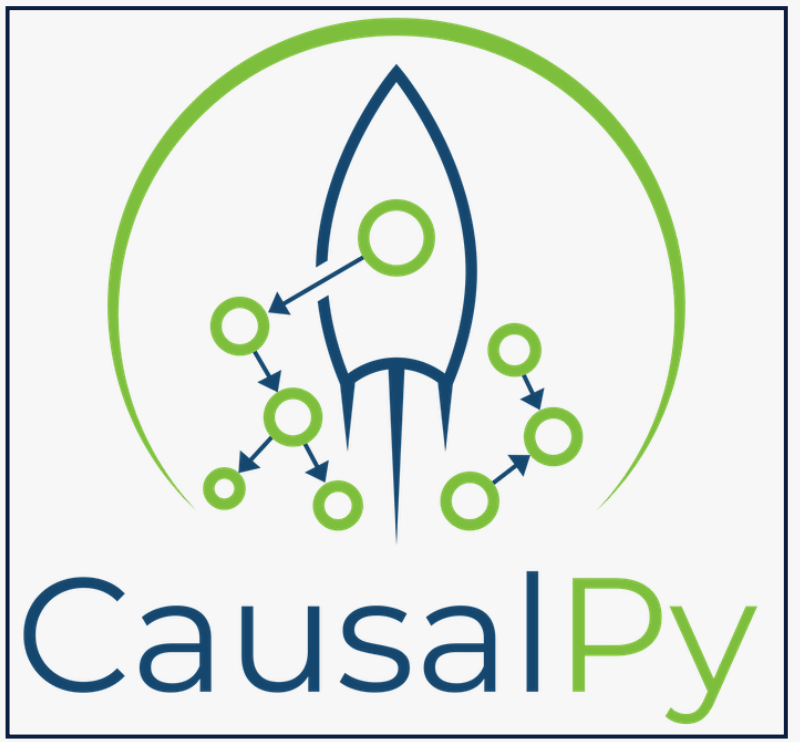
Application: Marketing Mix Modeling

Marketing Mix Modeling (MMM) is by nature a causal inference problem.

Question: How to optimally allocate the budget across channels to maximize the revenue in the next period?

We need to estimate marketing efficiency per channel (ROAS) to then optimize the budget allocation.

PyMC-Labs Ecosystem



Takeaways

What PPLs unlock for causal inference

- **Model = DAG.** Writing the generative process *is* stating your assumptions. → *Lalonde, Mediation*
- **Priors do real work.** Domain knowledge tightens estimates and rules out absurdity. → *A/B Testing, CUPED*
- **Uncertainty for free.** Posterior credible intervals on every causal estimand. → *all results slides*
- **Flexible likelihoods & structure.** Swap Normal → Gamma, add hierarchy, add latents. → *Gamma GLM, Mundlak, CEVAE*
- **do-operator built in.** Counterfactuals are a model transformation, not a separate library. → *ATE, Mediation*
- **Same machinery everywhere.** From the smallest A/B test to production MMM.

References — Posts & Packages

Blog Posts

- Introduction to Causal Inference with PPLs
- Fixed and Random Effects Models
- Causal Effect Estimation with Variational Inference and Latent Confounders
- Mediation Analysis and (In)Direct Effects with PyMC
- Prior Predictive Modeling in Bayesian A/B Testing
- Introduction to Bayesian Power Analysis
- Bayesian Power Analysis for A/B Testing
- Bayesian CUPED

Packages

- CausalPy
- ChiRho
- DoWhy
- NumPyro
- PyMC
- PyMC-Marketing

References — Books & Papers

Books

- Pearl, J. *Causality: Models, Reasoning and Inference*
- McElreath, R. *Statistical Rethinking* (2nd ed.)
- Facure, M. *Causal Inference for The Brave and True*
- Ness, R. O. *Causal AI* (Manning)

Papers

- Rubin, D. B. (1974). *Estimating causal effects of treatments in randomized and nonrandomized studies.*
- Mundlak, Y. (1978). *On the Pooling of Time Series and Cross Section Data.*
- Kruschke, J. K. & Liddell, T. M. (2018). *The Bayesian New Statistics: hypothesis testing, estimation, meta-analysis, and power analysis from a Bayesian perspective.*
- Louizos, C. et al. (2017). *Causal Effect Inference with Deep Latent-Variable Models* (NeurIPS).

Thank You!

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